

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396782
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Cosmescu; Ioan

Ultrapolar electrosurgery blade and ultrapolar electrosurgery blade assembly and method for making same

Abstract

An ultrapolar electrosurgery blade and an ultrapolar electrosurgery blade assembly and method for making same. The ultrapolar electrosurgery blade includes top and bottom thin elongated conductive members in vertical alignment and spaced apart from one another along their lengths where each of the top and bottom thin elongated conductive members includes opposing planar sides, a sharp cutting end, and an opposite non-cutting end, and a non-conductive coating or housing covering both opposing sides of the top and bottom thin elongated conductive members and the space located between them where the cutting ends of the elongated conductive members and their opposite non-cutting ends remain exposed. The ultrapolar electrosurgery blade assembly of the present invention having argon beam capability further includes a non-conductive tube member having a hollow tubular shaped opening with a slot where the slot is positioned over the top of the ultrapolar electrosurgery blade.

Inventors:	Cosmescu; Ioan (Phoenix, AZ)
Applicant:	I.C. Medical, Inc. (Phoenix, AZ)
Family ID:	1000008779053
Assignee:	I.C. Medical, Inc. (Phoenix, AZ)
Appl. No.:	18/679535
Filed:	May 31, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240315758 A1	Sep. 26, 2024

Related U.S. Application Data

continuation parent-doc US 17467327 20210906 US 11998261 child-doc US 18679535
continuation parent-doc US 15913569 20180306 US 11109907 20210907 child-doc US 17467327
us-provisional-application US 62576213 20171024
us-provisional-application US 62467739 20170306

Publication Classification

Int. Cl.: **A61B18/14** (20060101); **A61B18/00** (20060101); **A61B18/04** (20060101); **A61B18/12** (20060101)

U.S. Cl.:

CPC **A61B18/14** (20130101); **A61B18/042** (20130101); A61B2018/00589 (20130101); A61B2018/00601 (20130101); A61B2018/00607 (20130101); A61B2018/1253 (20130101); A61B2018/1412 (20130101); A61B2018/1467 (20130101)

Field of Classification Search

CPC: A61B (2018/00589); A61B (2018/00601); A61B (2018/00607); A61B (2018/1253); A61B (2018/126); A61B (2018/1412); A61B (2018/1415); A61B (2018/1467); A61B (18/042); A61B (18/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4248231	12/1980	Herczog et al.	N/A	N/A
4674498	12/1986	Stasz	N/A	N/A
4770067	12/1987	Liu et al.	N/A	N/A
4850353	12/1988	Stasz et al.	N/A	N/A
4958539	12/1989	Stasz et al.	N/A	N/A
5125927	12/1991	Belanger	N/A	N/A
5281216	12/1993	Klicek	N/A	N/A
5382247	12/1994	Cimino et al.	N/A	N/A
7033354	12/2005	Keppel	N/A	N/A
11109907	12/2020	Cosmescu	N/A	N/A
11166757	12/2020	Cosmescu	N/A	N/A
11903629	12/2023	Cosmescu	N/A	A61B 18/04
11998261	12/2023	Cosmescu	N/A	A61B 18/14
2003/0130655	12/2002	Woloszko et al.	N/A	N/A
2006/0241588	12/2005	Heim et al.	N/A	N/A
2010/0094283	12/2009	Cosmescu	N/A	N/A
2013/0110108	12/2012	Davison et al.	N/A	N/A
2013/0331657	12/2012	Basson et al.	N/A	N/A
2014/0257273	12/2013	Cosmescu	N/A	N/A
2016/0317209	12/2015	Cosmescu	N/A	N/A
2017/0056054	12/2016	Dickerson et al.	N/A	N/A
2017/0319255	12/2016	Cosmescu	N/A	N/A

2018/0071011	12/2017	Cosmescu	N/A	N/A
2022/0054182	12/2021	Cosmescu	N/A	N/A
2024/0189011	12/2023	Cosmescu	N/A	A61B 18/16

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0280798	12/1987	EP	N/A
2010-508124	12/2009	JP	N/A
2012-143276	12/2011	JP	N/A
2016-508847	12/2015	JP	N/A
9940858	12/1998	WO	N/A

Primary Examiner: Dvorak; Linda C

Assistant Examiner: Blaise; Bradford C.

Attorney, Agent or Firm: Zeman-Mullen & Ford, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of non-provisional patent application having Ser. No. 17/467,327, issued as U.S. Pat. No. 11,998,261 on Jun. 4, 2024, which is a continuation application of non-provisional patent application having Ser. No. 15/913,569, issued as U.S. Pat. No. 11,109,907 on Sep. 7, 2021, which claims priority to provisional patent application having Ser. No. 62/467,739, filed Mar. 6, 2017, and provisional patent application having Ser. No. 62/576,213, filed Oct. 24, 2017, which are herein incorporated by reference in their entireties.

FIELD OF INVENTION

(1) The present invention is generally directed to an ultrapolar electrosurgery blade and an ultrapolar electrosurgery blade assembly which use monopolar energy in a bipolar mode for cutting and coagulation. The ultrapolar electrosurgery blade includes top and bottom thin elongated conductive members in vertical alignment with one another and spaced apart from one another along their lengths wherein each of the top and bottom thin elongated conductive members includes opposing planar sides, a sharp cutting end, and an opposite non-cutting end, and a non-conductive coating covering both opposing sides of the top and bottom thin elongated conductive members and the space located between the top and bottom thin elongated conductive members wherein at least a portion of the cutting ends of the top and bottom thin elongated conductive members and their opposite non-cutting ends remain exposed. During use, one of the top and bottom thin elongated conductive members functions as an active electrode while the other thin elongated conductive member functions as a return electrode. The ultrapolar electrosurgery blade may further include a non-conductive support member/socket having two openings therein in vertical alignment with one another wherein a portion of the top and bottom thin elongated conductive members located near their non-cutting ends are respectively contained within one of the two openings of the support member/socket so that the ultrapolar electrosurgery blade of the present invention can be seated and retained within an electrosurgery pencil. The ultrapolar electrosurgery blade of the present invention is capable of cutting tissue with the sharp conductive cutting ends of the blade without using RF energy as well as cutting tissue with the sharp non-conductive cutting end/edge that is located between the sharp conductive cutting ends. In addition, the ultrapolar electrosurgery blade of the present invention is capable of coagulating tissue and/or enhanced cutting of tissue by

supplying low power to the ultrapolar electrosurgery blade, and simultaneously cutting and coagulating tissue by cutting tissue with the sharp cutting ends of the ultrapolar electrosurgery blade while coagulating tissue by applying low power to the ultrapolar electrosurgery blade.

(2) The present invention is also directed to an ultrapolar electrosurgery blade assembly with argon beam capability which includes the previously described ultrapolar electrosurgery blade, a non-conductive tube member having a hollow tubular shaped opening contained therein and a slot where the slot is positioned over the top of the ultrapolar electrosurgery blade, and a conductive hollow tubular member contained within at least a portion of the non-conductive tube member. Conductive projections may extend from the conductive hollow tubular member and/or a portion of the top thin elongated conductive member that is exposed depending on the type of coagulation to be performed such as argon plasma coagulation alone and/or argon plasma assisted coagulation. The non-conductive coating is a continuous coating that also fills any space located between the sharp cutting ends of the top and bottom thin elongated conductive members to create a sharp non-conductive cutting end of the ultrapolar electrosurgery blade located between the sharp conductive cutting ends of the top and bottom thin elongated conductive members. The conductive hollow tubular member contained within the non-conductive tube member may also include a slot that is positioned over a top portion of the ultrapolar electrosurgery blade. The ultrapolar electrosurgery blade assembly with argon beam capability provides argon plasma coagulation of tissue and/or argon plasma assisted cutting and/or argon plasma assisted coagulation of tissue depending on the location and configuration of the active and return electrodes of the ultrapolar electrosurgery blade.

BACKGROUND OF THE INVENTION

(3) Electrosurgery uses an RF electrosurgical generator (also known as an electrosurgical unit or ESU) and a handpiece with an electrode to provide high frequency, alternating radio frequency (RF) current input at various voltages to cut or coagulate biological tissue. The handpiece may be a monopolar instrument with one electrode or a bipolar instrument with two electrodes. When using a monopolar instrument, a return electrode pad is attached to the patient and the high frequency electrical current flows from the generator, to the monopolar instrument, through the patient to the patient return electrode pad, and back to the generator. Monopolar electrosurgery is commonly used due to its versatility and effectiveness. However, the excessive heat generated with monopolar electrosurgery can cause excessive tissue damage and necrosis of the tissue because the return electrode positioned on the back of the patient causes high voltage and high RF energy to pass through the patient.

(4) In bipolar electrosurgery, active output and patient return functions both occur at the surgery site because both the active and return electrodes are contained in the bipolar instrument. Therefore, the path of the electrical current is confined to the biological tissue located between the active and return electrodes. Although bipolar electrosurgery enables the use of lower voltages and less energy and thereby reduces or eliminates the likelihood of tissue damage and sparking associated with monopolar electrosurgery, it has limited ability to cut and coagulate large bleeding areas.

(5) Since surgical tools and devices currently available to surgeons require switching between cutting and coagulation modes during the surgical procedure, there is a need for a surgical device or tool that enables a surgeon or user to utilize the best methods used for cutting and cessation of bleeding at the surgical site at the same time, or simultaneously, in addition to being able to use them separately. An electrosurgery blade having a sharp edge for cutting and RF for coagulation would meet this need. The ultrapolar electrosurgery blade of the present invention which uses monopolar energy in a bipolar mode has sharp cutting edges made of a hard conductive material, such as stainless steel, tungsten, etc. that are separated by a sharp non-conductive cutting edge that can all be used for precisely cutting tissue without the use of any RF energy. However, RF energy can also be used with the ultrapolar electrosurgery blade of the present invention for coagulation. When low voltage is used to supply power to the ultrapolar electrosurgery blade of the present

invention for coagulation, the sharp cutting edges of the ultrapolar electrosurgery blade can simultaneously be used for cutting without the need to provide higher voltage to the ultrapolar electrosurgery blade to carry out the cutting. Therefore, there is no need to switch over to a cutting mode to perform cutting and instead both cutting and coagulation can be performed simultaneously at low power levels supplied from the generator.

(6) Moreover, the low power used to employ the ultrapolar electrosurgery blade of the present invention for both cutting and coagulation substantially reduces the damage to the lateral tissue and the tissue will not stick to the ultrapolar blade. Further, since the ultrapolar electrosurgery blade of the present invention includes top and bottom conductive members/electrodes that are both attached to the generator, only a very small amount of a patient's tissue located between the electrodes or adjacent to the electrodes is included in the circuit thereby eliminating the risk of current diversion to other parts of the patient that can occur in monopolar systems where the entire patient is in the circuit.

(7) It is also common to use argon beam coagulators during electrosurgery. In argon beam coagulation (ABC), plasma is applied to tissue by a directed beam of ionized argon gas (plasma) which causes a uniform and shallow coagulation surface thereby stopping blood loss. In some instances, electrosurgery is often the best method for cutting and argon beam coagulation is often the best method for cessation of bleeding during surgery. Surgeons typically need to switch between argon beam coagulation and electrosurgery modes depending on what is happening during the surgery and what they need to achieve at a particular point in the surgery such as making incisions in tissue by cutting, or stopping the bleeding at the surgical site.

(8) Since surgical tools and devices currently available to surgeons require switching between cutting and argon beam coagulation during the surgical procedure, there is a need for a surgical device or tool that enables a surgeon or user to utilize the best methods used for cutting and cessation of bleeding at the surgical site at the same time, or simultaneously, in addition to being able to use them separately. An ultrapolar electrosurgery blade having a sharp edge for cutting and argon beam capability for capsulation would meet this need. The ultrapolar electrosurgery blade assembly of the present invention with a sharp cutting edge and argon beam capability enables a user or surgeon to perform cutting and coagulation without the need to switch between cutting and coagulation modes. There is also a need for an electrosurgical device that enables a user or surgeon to choose from a number of different separate or combined tissue cutting and coagulation methods since different methods may work best depending on the surgical procedure and circumstances that present themselves during surgery.

(9) The ultrapolar electrosurgery blade assembly with argon beam capability of the present invention is capable of coagulating a patient's tissue using argon plasma alone without contacting the patient's tissue (i.e. non-contact argon beam coagulation). In this embodiment of the ultrapolar electrosurgery blade assembly, an exposed portion of the return electrode of the ultrapolar electrosurgery blade is positioned near the top of the electrosurgery blade such that it is in alignment with the conductive hollow tubular member through which the argon gas is introduced and the conductive projection extending from an end of the conductive tube member so that a complete circuit is formed to ionize the argon gas for argon plasma coagulation. The ultrapolar electrosurgery blade assembly of the present invention is also capable of cutting a patient's tissue using the sharp cutting edge (comprising both conductive and non-conductive materials) of the ultrapolar blade alone without any use of RF energy and without any use of argon plasma. The ultrapolar electrosurgery blade assembly of the present invention can also enhance the cutting of a patient's tissue using the sharp conductive cutting edges of the ultrapolar blade by also supplying RF energy to the ultrapolar electrosurgery blade. Moreover, the ultrapolar electrosurgery blade assembly of the present invention having a sharp cutting edge and argon beam capability enables a user or surgeon to simultaneously perform cutting and coagulation without the need to switch between cutting and coagulation modes by performing argon plasma assisted cutting and

coagulation. For example, the sharp cutting edge of the ultrapolar blade can be used without any RF energy for cutting while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue. In another example, low power may be applied to the ultrapolar blade to coagulate tissue or enhance cutting of tissue while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue.

(10) Both the ultrapolar electrosurgery blade of the present invention and the ultrapolar electrosurgery blade assembly having argon beam capability of the present invention could be used with an electrosurgery handpiece/pencil with smoke evacuation capability or an electrosurgery handpiece/pencil without smoke evacuation capability. Both the ultrapolar electrosurgery blade of the present invention and the ultrapolar electrosurgery blade assembly having argon beam capability of the present invention enable the surgeon or user to increase both the efficiency and accuracy of the surgery by enabling the surgeon or user to perform different methods of cutting and coagulating tissue either separately or simultaneously. In instances where tissue cutting and coagulation are performed at the same time without switching between modes or methods, operating time is decreased and the lateral damage to the tissue is reduced or eliminated. Further, use of monopolar energy in a bipolar mode with the ultrapolar electrosurgery blade of the present invention and the ultrapolar electrosurgery blade assembly with argon beam capability of the present invention essentially eliminates the risk of current diversion that can occur in monopolar systems. In addition, performing both tissue cutting and coagulation at the same time along with smoke evacuation will protect the surgeon and staff from inhaling smoke and particles. It will also enable the surgeon or user to more clearly view the surgical site to ensure accuracy during the procedure without the need to stop and switch modes in order to stop bleeding at the surgery site before being able to clearly see the surgical site.

SUMMARY OF THE INVENTION

(11) The present invention is directed to an ultrapolar electrosurgery blade which uses monopolar energy in a bipolar mode and which includes top and bottom thin elongated conductive members in vertical alignment with one another and spaced apart from one another along their lengths wherein each of the top and bottom thin elongated conductive members includes opposing planar sides, a sharp cutting end for cutting, and an opposite non-cutting end, and a non-conductive coating covering both opposing planar sides of the top and bottom thin elongated conductive members and the space located between them wherein the cutting ends of the thin elongated conductive members and their opposite non-cutting ends remain exposed. During use, one of the top and bottom thin elongated conductive members functions as an active electrode while the other thin elongated conductive member functions as a return electrode. The ultrapolar electrosurgery blade may further include a non-conductive support member/socket having two openings therein positioned in vertical alignment with one another wherein a portion of each of the top and bottom thin elongated conductive members located near their non-cutting ends are respectively contained within one of the two openings of the support member/socket so that the ultrapolar electrosurgery blade of the present invention can be seated and retained within an electrosurgery pencil.

(12) The top and bottom thin elongated conductive members may be formed from a single thin conductive member having vertically aligned top and bottom elongated conductive members spaced apart from one another along their lengths with each having a separate sharp cutting end at one end and a non-cutting end at their opposite ends where their non-cutting ends are joined. The non-conductive coating may then be applied to the single thin conductive member (which includes the top and bottom elongated conductive members and the space located between the top and bottom elongated conductive members) to form an electrosurgery blade where at least a portion of the cutting ends of the top and bottom elongated conductive members and their joined opposing non-cutting ends remain exposed and not covered by the non-conductive coating. The joined non-

cutting ends of the top and bottom elongated conductive members can then be removed to produce separately exposed and unconnected non-cutting ends for the top and bottom elongated conductive members which can be respectively inserted into a non-conductive support member/socket having two openings (as described above). One advantage in forming the ultrapolar electrosurgery blade of the present invention using a single thin conductive member having vertically aligned top and bottom elongated conductive members spaced apart from one another along their lengths with separate sharp cutting ends at one end and joined opposite non-cutting ends where the joined ends are later removed to produce separate non-cutting ends is that it facilitates the construction and production of the ultrapolar electrosurgery blade by providing a unitary component for creating separate elements of the blade thereby increasing the consistency and accuracy of the blades. Another advantage of this type of formation of the ultrapolar electrosurgery blades are the increased efficiencies in the production of the blades and the reduction in production costs. Still another advantage of this type of blade formation for the ultrapolar electrosurgery blade of the present invention is that it enhances the strength of the blade as well as the proper functioning of the blade.

(13) In one exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, the non-conductive coating covers at least a portion of the top of the top thin elongated conductive member and at least a portion of the bottom of the bottom thin elongated conductive member. The non-conductive coating may be a continuous coating that also fills in any space located between the sharp cutting ends of the top and bottom thin elongated conductive members. In another exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, a portion of the top of the top thin elongated conductive member is exposed between portions of the non-conductive coating located on the top of the electrosurgery blade and a portion of the bottom of the bottom thin elongated conductive member is exposed between portions of the non-conductive coating located on the bottom of the electrosurgery blade. The ultrapolar electrosurgery blade of the present invention may have a sharp cutting edge that is comprised of the sharp cutting ends of the top and bottom thin elongated conductive members separated by a sharp non-cutting end comprised of the non-conductive coating.

(14) The top and bottom thin elongated conductive members (as well as the single thin conductive member that the top and bottom elongated members may be formed from) may comprise a hard metal such as, for example, stainless steel, titanium, and/or tungsten. The non-conductive coating of the ultrapolar electrosurgery blade of the present invention and the non-conductive support member may be comprised of a ceramic material.

(15) The ultrapolar electrosurgery blade of the present invention which uses monopolar energy in a bipolar mode has sharp cutting edges made of a hard conductive material, such as stainless steel, tungsten, etc. that are separated by a sharp non-conductive cutting edge that can all be used for precisely cutting tissue without the use of any RF energy. However, RF energy can also be used with the ultrapolar electrosurgery blade of the present invention for coagulation. When low voltage is used to supply power to the ultrapolar electrosurgery blade of the present invention for coagulation, the sharp cutting edges of the ultrapolar electrosurgery blade can simultaneously be used for cutting without the need to provide higher voltage to the ultrapolar electrosurgery blade to carry out the cutting. Therefore, there is no need to switch over to a cutting mode to perform cutting and instead both cutting and coagulation can be performed simultaneously at low power levels supplied from the generator.

(16) The ultrapolar electrosurgery blade assembly of the present invention having argon beam capability includes the ultrapolar electrosurgery blade described above and further includes a non-conductive tube member having a hollow tubular shaped opening contained therein and a slot where the slot is positioned over the top of the ultrapolar electrosurgery blade, and a conductive hollow tubular member contained within at least a portion of the non-conductive tube member. In one exemplary embodiment of the ultrapolar electrosurgery blade assembly of the present

invention, a portion of the top of the top thin elongated conductive member is exposed between portions of the non-conductive coating located on the top of the ultrapolar electrosurgery blade and contained within the non-conductive tube member and the ultrapolar electrosurgery blade assembly further includes a conductive projection extending from the conductive hollow tubular member contained within the non-conductive tube member and/or the portion of the top of the top thin elongated conductive member located on top of the electrosurgery blade and contained within the non-conductive tube member. In another exemplary embodiment of the ultrapolar electrosurgery blade assembly of the present invention, the non-conductive coating covers the top of the top thin elongated conductive member located between the conductive hollow tubular member and the exposed cutting end of the top thin elongated conductive member and the ultrapolar electrosurgery blade assembly further includes a conductive projection extending from an end of the conductive hollow tubular member contained within the non-conductive tube member.

(17) The conductive hollow tubular member contained within the non-conductive tube member may include a slot that, like the slot in the non-conductive tube member, is also positioned over at least a portion of the top of the ultrapolar electrosurgery blade. Like the top and bottom thin elongated conductive members of the ultrapolar electrosurgery blade, the conductive hollow tubular member, as well as the conductive projections, may comprise a hard metal such as, for example, stainless steel, titanium, and/or tungsten. Further, like the non-conductive coating of the ultrapolar electrosurgery blade, the non-conductive tube member may be comprised of a ceramic material.

(18) The ultrapolar electrosurgery blade assembly with argon beam capability of the present invention is capable of coagulating a patient's tissue using argon plasma alone without contacting the patient's tissue (i.e. non-contact argon beam coagulation). In this embodiment of the ultrapolar electrosurgery blade assembly, an exposed portion of the return electrode of the ultrapolar electrosurgery blade is positioned near the top of the electrosurgery blade such that it is in alignment with the conductive hollow tubular member through which the argon gas is introduced and the conductive projection extending from an end of the conductive tube member so that a complete circuit is formed to ionize the argon gas for argon plasma coagulation. The ultrapolar electrosurgery blade assembly of the present invention is also capable of cutting a patient's tissue using the sharp cutting edge (comprising both conductive and non-conductive materials) of the ultrapolar blade alone without any use of RF energy and without any use of argon plasma. The ultrapolar electrosurgery blade assembly of the present invention can also enhance the cutting of a patient's tissue using the sharp conductive cutting edges of the ultrapolar blade by also supplying RF energy to the ultrapolar electrosurgery blade. Moreover, the ultrapolar electrosurgery blade assembly of the present invention having a sharp cutting edge and argon beam capability enables a user or surgeon to simultaneously perform cutting and coagulation without the need to switch between cutting and coagulation modes by performing argon plasma assisted cutting and coagulation. For example, the sharp cutting edge of the ultrapolar blade can be used without any RF energy for cutting while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue. In another example, low power may be applied to the ultrapolar blade to coagulate tissue or enhance cutting of tissue while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view of an exemplary embodiment of a thin conductive member having top and

bottom thin elongated conductive members used to make the ultrapolar electrosurgery blade of the present invention;

(2) FIG. 2 is a side view of another exemplary embodiment of a thin conductive member having top and bottom thin elongated conductive members used to make the ultrapolar electrosurgery blade of the present invention;

(3) FIG. 3 is side perspective view of an exemplary embodiment of the ultrapolar electrosurgery blade of the present invention showing the exemplary embodiment of the thin conductive member of FIG. 1 coated with a non-conductive coating except for the cutting ends and the joined non-cutting ends of the top and bottom elongated conductive members where the non-conductive coating is represented by light shade hash marks and/or hash marks made of unconnected dots;

(4) FIG. 4 is a front end view of a support member/socket/connector member which retains a portion of the unconnected non-cutting ends of the top and bottom elongated conductive members of the ultrapolar electrosurgery blade thereby facilitating the connection of the ultrapolar electrosurgery blade of the present invention to an electrosurgery pencil.

(5) FIG. 5 is a top view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 3 with the thin conductive member shown in phantom;

(6) FIG. 6 is a bottom view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 3 with the thin conductive member shown in phantom;

(7) FIG. 7 is an external side view showing the exemplary embodiment of the ultrapolar electrosurgery blade shown in FIG. 3 with the joined portion of the non-cutting ends of the top and bottom elongated conductive members removed and the top and bottom elongated conductive members covered by the non-conductive coating shown in phantom;

(8) FIG. 8 is a top view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 7 with the top elongated conductive member covered by the non-conductive coating shown in phantom;

(9) FIG. 9 is a bottom view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 7 with the bottom elongated conductive member covered by the non-conductive coating shown in phantom;

(10) FIG. 10 is a front end view of an exemplary embodiment of a support member/connector member into which unconnected non-cutting ends of the top and bottom elongated conductive members of the ultrapolar electrosurgery blade are placed so that the ultrapolar electrosurgery blade of the present invention can be connected to, and unconnected or removed from, an electrosurgery pencil;

(11) FIG. 11 is an end view of the support member/connector member shown in FIG. 10 showing conductive unconnected non-cutting ends of the ultrapolar electrosurgery blade of the present invention retained within the openings in the support member/connector member;

(12) FIG. 12 is a partial top view of another exemplary embodiment of the ultrapolar electrosurgery blade of the present invention showing a sharp cutting end beveled on both sides to create a sharp cutting tip;

(13) FIG. 13 is a partial perspective view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIGS. 3 and 7;

(14) FIGS. 14 and 15 are opposing perspective side views of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 7 to further reveal the shape of the ultrapolar electrosurgery blade of the present invention;

(15) FIGS. 16-17 show different views of an exemplary non-conductive support member/socket/connector member that comprises part of the ultrapolar electrosurgery blade of the present invention when used in a non-telescopic electrosurgery pencil;

(16) FIGS. 18-19 show different views of an exemplary non-conductive support member/socket/connector member that comprises part of the ultrapolar electrosurgery blade of the present invention when used in a telescopic electrosurgery pencil;

(17) FIG. **20** is a partial perspective view of an exemplary embodiment of the ultrapolar electrosurgery blade assembly of the present invention having argon beam capability for providing argon plasma assisted coagulation;

(18) FIG. **21** is a side perspective view of another exemplary embodiment of the ultrapolar electrosurgery blade assembly of the present invention having argon beam capability for providing argon plasma assisted coagulation with the return electrode extending along part of the bottom of the ultrapolar blade; and

(19) FIG. **22** is a side perspective view of still another exemplary embodiment of the ultrapolar electrosurgery blade assembly of the present invention having argon beam capability which is capable of providing both argon plasma coagulation and argon plasma assisted coagulation.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(20) The exemplary embodiments of the ultrapolar electrosurgery blade and ultrapolar electrosurgery blade assembly having argon beam capability of the present invention enable the surgeon or user to increase both the efficiency and accuracy of the surgery by enabling the surgeon or user to perform different methods of cutting and coagulating tissue either separately or simultaneously. The ultrapolar electrosurgery blade of the present invention is capable of cutting tissue with the sharp conductive cutting ends of the blade without using RF energy as well as cutting tissue with the sharp non-conductive cutting end/edge that is located between the sharp conductive cutting ends. In addition, the ultrapolar electrosurgery blade of the present invention is capable of coagulating tissue and/or enhanced cutting of tissue by supplying very low power, such as 5 to 15 watts, to the ultrapolar electrosurgery blade, and simultaneously cutting and coagulating tissue by cutting tissue with the sharp cutting ends of the ultrapolar electrosurgery blade while coagulating tissue by applying very low power to the ultrapolar electrosurgery blade.

(21) The ultrapolar electrosurgery blade assembly of the present invention with a sharp cutting edge and argon beam capability enables a user or surgeon to perform cutting and coagulation without the need to switch between cutting and coagulation modes. There is also a need for an electrosurgical device that enables a user or surgeon to choose from a number of different separate or combined tissue cutting and coagulation methods since different methods may work best depending on the surgical procedure and circumstances that present themselves during surgery. The ultrapolar electrosurgery blade assembly with argon beam capability of the present invention is capable of coagulating a patient's tissue using argon plasma alone without contacting the patient's tissue (i.e. non-contact argon beam coagulation). The ultrapolar electrosurgery blade assembly of the present invention is also capable of cutting a patient's tissue using the sharp cutting edge (comprising both conductive and non-conductive materials) of the ultrapolar blade alone without any use of RF energy and without any use of argon plasma. The ultrapolar electrosurgery blade assembly of the present invention can also enhance the cutting of a patient's tissue using the sharp conductive cutting edges of the ultrapolar blade by also supplying RF energy to the ultrapolar electrosurgery blade. Moreover, the ultrapolar electrosurgery blade assembly of the present invention having a sharp cutting edge and argon beam capability enables a user or surgeon to simultaneously perform cutting and coagulation without the need to switch between cutting and coagulation modes by performing argon plasma assisted cutting and coagulation. For example, the sharp cutting edge of the ultrapolar blade can be used without any RF energy for cutting while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue. In another example, low power may be applied to the ultrapolar blade to coagulate tissue or enhance cutting of tissue while the conductive tube through which the argon gas is introduced, and which is contained within the non-conductive tube, is activated and directed to provide ionized argon gas for argon plasma coagulation of tissue.

(22) The identity of the elements/features that relate to the numbers shown in the drawing figures are as follows: **10** ultrapolar electrosurgery blade **11** thin conductive member **12** top thin elongated

conductive member **14** bottom thin elongated conductive member **16** elongated space between top and bottom thin elongated conductive members **18** opposing planar sides (of top and bottom thin elongated conductive members) **22** sharp cutting end of top thin elongated conductive member **24** sharp cutting end of bottom thin elongated conductive member **26** opposite non-cutting end of top thin elongated conductive member **28** opposite non-cutting end of bottom thin elongated conductive member **30** portion of thin conductive member joining opposite non-cutting ends of top and bottom thin elongated conductive members **26** and **28** **31** thin conductive member **32** top thin elongated conductive member **34** bottom thin elongated conductive member **36** elongated space between top and bottom thin elongated conductive members **38** opposing planar sides (of top and bottom thin elongated conductive members) **42** sharp cutting end of top thin elongated conductive member **44** sharp cutting end of bottom thin elongated conductive member **46** opposite non-cutting end of top thin elongated conductive member **48** opposite non-cutting end of bottom thin elongated conductive member **50** portion of thin conductive member joining opposite non-cutting ends of top and bottom thin elongated conductive members **46** and **48** **60** non-conductive coating/housing **62** non-conductive support member/socket/connecting member **63** rounded top portion (of non-conductive support member/socket/connecting member) **65** rounded bottom portion (of non-conductive support member/socket/connecting member) **64** two vertically aligned openings **66** top of the top thin elongated conductive member **68** bottom of the bottom thin elongated conductive member **70** sharp non-conductive cutting end **72** non-conductive support member/socket/connecting member for ultrapolar telescopic electrosurgery pencil **73** rounded top portion (of non-conductive support member/socket/connecting member for ultrapolar telescopic electrosurgery pencil) **74** two vertically aligned openings **100** ultrapolar electrosurgery blade assembly **120** non-conductive tube member **122** hollow tubular shaped opening (of non-conductive tube member) **124** slot (of non-conductive tube member) **130** conductive hollow tubular member **132** conductive projection **200** ultrapolar electrosurgery blade assembly **220** non-conductive tube member **222** hollow tubular shaped opening (of non-conductive tube member) **224** slot (of non-conductive tube member) **230** conductive hollow tubular member **232** conductive projection **300** ultrapolar electrosurgery blade assembly **320** non-conductive tube member **322** hollow tubular shaped opening (of non-conductive tube member) **324** slot (of non-conductive tube member) **330** conductive hollow tubular member **332** conductive projection **334** slot (of conductive hollow tubular member)

(23) FIG. 1 is a side view of an exemplary embodiment of a thin conductive member **11** having top and bottom thin elongated conductive members **12**, **14** used to make the ultrapolar electrosurgery blade **10** of the present invention. Thin conductive member **11** includes a top thin elongated conductive member **12** and a bottom thin elongated conductive member **14** in vertical alignment with one another and separated from one another along their lengths by a space **16**. The width of the space **16** located between the top thin elongated conductive member **12** and the bottom thin elongated conductive member **14** is the same along more than half the length of the top and bottom thin elongated conductive members **12**, **14**. The top and bottom elongated conductive members **12**, **14** each have opposing planar sides **18**, with each opposing planar side **18** having a length and a width, a sharp cutting end **22**, **24** and an opposite non-cutting end **26**, **28** where the opposite non-cutting ends **26**, **28** are joined by a portion **30** of the thin conductive member **11**. The width of the space **16** that is the same along more than half the length of the top and bottom thin elongated conductive members **12**, **14** is smaller than the width of each of the opposing planar sides **18** of the top and bottom thin elongated conductive members **12**, **14** located adjacent to the sharp cutting ends **22**, **24**. The width of the space **16** that is the same along more than half the length of the top and bottom thin elongated conductive members **12**, **14** is also smaller than the width of each of the opposing planar sides **18** located at a middle of the length of the top and bottom thin elongated conductive members **12**, **14**. In one exemplary embodiment of the thin conductive member **11**, the sharp cutting ends **22**, **24** of the thin conductive member **11** form an angle X relative to a plane that

is in horizontal alignment with the bottom of the bottom thin elongated conductive member **14** where X is a sixty degree angle.

(24) A side view of another exemplary embodiment of a thin conductive member **31** having top and bottom thin elongated conductive members **32**, **34** used to make the ultrapolar electrosurgery blade **10** of the present invention is shown in FIG. 2. Like the thin conductive member **11** shown in FIG. 1, thin conductive member **31** includes a top thin elongated conductive member **32** and a bottom thin elongated conductive member **34** in vertical alignment with one another and separated from one another along their lengths by a space **36**. The width of the space **36** located between the top thin elongated conductive member **32** and the bottom thin elongated conductive member **34** is the same along more than half the length of the top and bottom thin elongated conductive members **32**, **34**. The top and bottom elongated conductive members **32**, **34** each have opposing planar sides **38**, with each opposing planar side **38** having a length and a width, a sharp cutting end **42**, **44** and an opposite non-cutting end **46**, **48** where the opposite non-cutting ends **46**, **48** are joined by a portion **50** of the thin conductive member **31**. The width of the space **36** that is the same along more than half the length of the top and bottom thin elongated conductive members **32**, **34** is smaller than the width of each of the opposing planar sides **38** of the top and bottom thin elongated conductive members **32**, **34** located adjacent to the sharp cutting ends **42**, **44**. The width of the space **36** that is the same along more than half the length of the top and bottom thin elongated conductive members **32**, **24** is also smaller than the width of each of the opposing planar sides **38** located at a middle of the length of the top and bottom thin elongated conductive members **32**, **34**. As shown in FIG. 2, the sharp cutting end **42** of top thin elongated conductive member **32** extends well beyond the sharp cutting end **44** of the bottom thin elongated conductive member **34** and the angle of the sharp cutting end **44** in relation to the bottom of the bottom thin elongated conductive member **34** is much steeper than the angle of the sharp cutting end **42** in relation to the bottom of the top thin elongated conductive member **32**. It will be understood by those skilled in the art that the sharp cutting ends of the top and bottom thin elongated conductive members of the ultrapolar electrosurgery blade may include any number of shapes and/or configurations depending on the type and circumstances of the surgical procedure to be performed using the ultrapolar electrosurgery blade.

(25) FIG. 3 is side perspective view of an exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention showing the exemplary embodiment of the thin conductive member **11** of FIG. 1 coated with a non-conductive coating **60** except for the cutting ends **22**, **24** and the joined non-cutting ends **26**, **28**, **30** of the top and bottom elongated conductive members **12**, **14** where the non-conductive coating **60** is represented by light shade hash marks and/or hash marks made of unconnected dots. FIG. 5 is a top view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. 3 with the thin conductive member **11** shown in phantom and FIG. 6 is a bottom view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. 3 with the thin conductive member **11** shown in phantom. As can be seen from FIGS. 3 and 5-6, the non-conductive coating **60** covers the thin conductive member **11** except for sharp cutting ends **22**, **24** of the top and bottom elongate conductive members **12**, **14**, a portion of the top **66** of the of the top elongated conductive member **12**, a portion of the bottom **68** of the bottom elongated conductive member **14**, the non-cutting ends **26**, **28** of the top and bottom elongated conductive members **12**, **14** and the portion **30** of the thin conductive member **11** that joins the non-cutting ends **26**, **28**.

(26) After the non-conductive coating **60** is applied to the thin conductive member **11** and the coating **60** is set, the portion **30** that joins the non-cutting ends **26**, **28** is removed as shown in FIG. 7 to provide an ultrapolar electrosurgery blade **10** having unconnected conductive non-cutting ends **26**, **28** supported by a support member/socket/connecting member **64** which facilitates connection of the ultrapolar electrosurgery blade **10** of the present invention to an electrosurgery pencil. FIG. 7 is an external side view showing the exemplary embodiment of the ultrapolar electrosurgery blade

10 shown in FIG. **3** with the joined portion **30** of the non-cutting ends **26, 28** of the top and bottom elongated conductive members **12, 14** removed and most of the top and bottom elongated conductive members **12, 14** covered by the non-conductive coating **60** shown in phantom. Advantages in forming the ultrapolar electrosurgery blade **10** of the present invention using a single thin conductive member **11** having vertically aligned top and bottom elongated conductive members **12, 14** spaced apart from one another along their lengths with separate sharp cutting ends **22, 24** at one end and joined opposite non-cutting ends **26, 28, 30** where the joined ends are later removed to produce separate non-cutting ends **26, 28** include 1) facilitation of the construction and production of the ultrapolar electrosurgery blade of the present invention by providing a unitary component for creating separate elements of the blade thereby increasing the consistency and accuracy of the blades, 2) increased efficiencies in the production of the blades and the reduction in production costs, and 3) enhanced strength of the blade as well as the enhanced proper functioning of the blade.

(27) FIG. **8** is a top view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. **7** with the top elongated conductive member **12** that is covered by the non-conductive coating **60** shown in phantom. A portion of the top **66** of the top elongated conductive member **12** is exposed between portions of non-conductive coating **60** located on a top of the ultrapolar electrosurgery blade. FIG. **9** is a bottom view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. **7** with the bottom elongated conductive member **14** covered by the non-conductive coating **60** shown in phantom. A portion of the bottom **68** of the bottom elongated conductive member **14** is exposed between portions of non-conductive coating **60** located on a bottom of the ultrapolar electrosurgery blade.

(28) Further, as shown in FIG. **7**, the non-conductive coating is a continuous coating that fills elongated space **16** located between the top and bottom elongated conductive members **12, 14** as well as any space located between the sharp cutting ends **22, 24** of the top and bottom elongated conductive members **12, 14**. The space between the sharp cutting ends **22, 24** of the top and bottom elongated conductive members **12, 14** that is filled with the non-conductive coating **60** forms a sharp non-conductive cutting end positioned between the sharp conductive cutting ends **22, 24** of the ultrapolar electrosurgery pencil **10**.

(29) FIG. **10** is a front end view of an exemplary embodiment of a support member/socket/connector member **62** into which unconnected non-cutting ends **26, 28** of the top and bottom elongated conductive members **12, 14** of the ultrapolar electrosurgery blade **10** are placed so that the ultrapolar electrosurgery blade **10** of the present invention can be easily connected to, and unconnected or removed from, an electrosurgery pencil. The support member/socket/connector member **62** includes two vertically aligned openings **64** so that conductive non-cutting ends **26, 28** can be respectively retained in them. An end view of the support member/connector member **62** shown in FIG. **10** showing conductive unconnected non-cutting ends **26, 28** of the ultrapolar electrosurgery blade **10** of the present invention retained within the openings **64** in the support member/socket/connector member **62** is shown in FIG. **11**.

(30) A partial top view of another exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention showing a sharp cutting end beveled on both sides to create a sharp cutting tip **22** is shown in FIG. **12**. A partial perspective view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIGS. **3** and **7** is depicted in FIG. **13**. FIG. **13** clearly shows sharp conductive cutting ends **22, 24** of the top and bottom thin elongated conductive members **12, 14** exposed after coating the top and bottom thin elongated conductive members **12, 14** with non-conductive coating **60** and sharp non-conductive cutting end **74** comprised of the non-conductive coating **60** located between the sharp conductive cutting ends **22, 24**. FIG. **13** also clearly shows a portion of the top **66** of the top elongated conductive member **12** exposed between portions of non-conductive coating **60** located on top of the ultrapolar electrosurgery blade **10** and in communication with the sharp cutting end **22** of the top elongated

conductive member **12**.

(31) FIGS. **14** and **15** are opposing perspective side views of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. **7** to further reveal the shape of the ultrapolar electrosurgery blade **10** of the present invention. As shown in FIGS. **14** and **15**, the conductive coating **60** which covers most of the top and bottom thin elongated conductive members **12**, **14** follows the shape of the top of the top elongated conductive member **12** and the shape of the bottom of the bottom elongated conductive member **14** until it gets to the portions of the top **66** of the top elongated conductive member **12** and the bottom **68** of the bottom elongated conductive member **14** that remain exposed between portions of non-conductive coating on the top of the ultrapolar electrosurgery blade **10** that are located adjacent to the cutting ends **22**, **24** of the top and bottom elongated conductive members **12**, **14**. In addition, as shown in FIGS. **14** and **15**, the non-conductive support member/socket/connecting member **62** is designed to retain non-cutting ends **26**, **28** of top and bottom elongated conductive members **12**, **14** and/or portions of the top and bottom elongated conductive members **12**, **14** located near the non-cutting ends **26**, **28** to facilitate connection of the ultrapolar electrosurgery blade **10** of the present invention with an electrosurgery pencil. The width of the non-conductive support member/socket/connecting member **62** is greater than a middle width of the ultrapolar electrosurgery blade **10** which is in turn greater than a width of the ultrapolar electrosurgery blade **10** located near the sharp cutting edge of the ultrapolar electrosurgery blade **10** which includes the sharp conductive cutting ends **22**, **24** separated by the sharp non-conductive cutting end **74**. The ultrapolar electrosurgery blade **10** has a three step configuration along the top and bottom lengths of the blade **10** which relate to the heights of the ultrapolar electrosurgery blade **10**.

(32) FIGS. **16-17** show different views of an exemplary non-conductive support member/socket/connector member that comprises part of the ultrapolar electrosurgery blade **10** of the present invention when used in a non-telescopic electrosurgery pencil and FIGS. **18-19** show different views of an exemplary embodiment of the non-conductive support member/socket/connector member that comprises part of the ultrapolar electrosurgery blade **10** of the present invention when used in a telescopic electrosurgery pencil. Non-conductive support member/socket/connecting member **62** includes a rounded top portion **63**, a rounded bottom portion **65**, and two vertically aligned openings **64** for receiving non-cutting ends **26**, **28** of top and bottom elongated conductive members **12**, **14** and/or portions of the top and bottom elongated conductive members **12**, **14** located near the non-cutting ends **26**, **28**. Non-conductive support member/socket/connecting member **72** includes a rounded top portion **73** and two vertically aligned openings **74** for receiving non-cutting ends **26**, **28** of top and bottom elongated conductive members **12**, **14** and/or portions of the top and bottom elongated conductive members **12**, **14** located near the non-cutting ends **26**, **28**.

(33) FIG. **20** is a partial perspective view of an exemplary embodiment of the ultrapolar electrosurgery blade assembly **100** of the present invention having argon beam capability for providing argon plasma assisted coagulation. Ultrapolar electrosurgery blade assembly **100** includes the ultrapolar electrosurgery blade **10** previously described above and further includes a non-conductive tube member **120** having a hollow tubular shaped opening **122** contained therein and a slot **124** where the slot is positioned over a top of the ultrapolar electrosurgery blade **10**. The ultrapolar electrosurgery blade assembly **100** further includes a conductive hollow tubular member **130** that is contained within at least a portion of the non-conductive tube member **120**. Conductive hollow tubular member **130** may also include a conductive projection **132**. The sharp cutting edge (comprising conductive cutting ends **22**, **26** separated by sharp non-conductive cutting end **74**) or a portion of the sharp cutting edge can be used without RF energy for cutting while argon gas is introduced through the conductive hollow tubular member **130** contained within the non-conductive tube member **120** while the conductive hollow tubular member **130** is activated and the conductive projection **132** can direct the ionized argon gas for argon plasma coagulation of tissue.

Alternatively, low power may be applied to the ultrapolar electrosurgery blade **10** to coagulate tissue or enhance cutting of tissue while argon gas is introduced through the conductive hollow tubular member **130** contained within the non-conductive tube member **120** while the conductive hollow tubular member **130** is activated and the conductive projection **132** can direct the ionized argon gas for argon plasma coagulation of tissue.

(34) FIG. **21** is a side perspective view of another exemplary embodiment of the ultrapolar electrosurgery blade assembly **200** of the present invention having argon beam capability for providing argon plasma assisted coagulation with the return electrode extending along part of the bottom of the ultrapolar blade. Ultrapolar electrosurgery blade assembly **200** includes the ultrapolar electrosurgery blade **10** previously described above and further includes a non-conductive tube member **220** having a hollow tubular shaped opening **222** contained therein and a slot **224** where the slot is positioned over a top of the ultrapolar electrosurgery blade **10**. The ultrapolar electrosurgery blade assembly **200** further includes a conductive hollow tubular member **230** that is contained within at least a portion of the non-conductive tube member **220**. Conductive hollow tubular member **230** may also include a conductive projection **232**. The sharp cutting edge (comprising conductive cutting ends **22**, **26** separated by sharp non-conductive cutting end **74**) or a portion of the sharp cutting edge can be used without RF energy for cutting while argon gas is introduced through the conductive hollow tubular member **230** contained within the non-conductive tube member **220** while the conductive hollow tubular member **230** is activated and the conductive projection **232** can direct the ionized argon gas for argon plasma coagulation of tissue. Alternatively, low power may be applied to the ultrapolar electrosurgery blade **10** to coagulate tissue or enhance cutting of tissue while argon gas is introduced through the conductive hollow tubular member **230** contained within the non-conductive tube member **220** while the conductive hollow tubular member **230** is activated and the conductive projection **232** can direct the ionized argon gas for argon plasma coagulation of tissue thereby employing argon plasma assisted cutting and/or coagulation

(35) FIG. **22** is a side perspective view of still another exemplary embodiment of the ultrapolar electrosurgery blade assembly **300** of the present invention having argon beam capability which is capable of providing both argon plasma coagulation and argon plasma assisted coagulation. Ultrapolar electrosurgery blade assembly **300** includes the ultrapolar electrosurgery blade **10** previously described above and further includes a non-conductive tube member **320** having a hollow tubular shaped opening **322** contained therein and a slot **324** where the slot is positioned over a top of the ultrapolar electrosurgery blade **10**. The ultrapolar electrosurgery blade assembly **300** further includes a conductive hollow tubular member **330** that is contained within at least a portion of the non-conductive tube member **320**. Conductive hollow tubular member **330** may also include a conductive projection **332**. In this embodiment of the ultrapolar electrosurgery blade assembly **300**, an exposed portion of the return electrode **22** of the ultrapolar electrosurgery blade **10** is positioned near the top of the electrosurgery blade **10** such that it is in alignment with the conductive hollow tubular member **330**, through which the argon gas is introduced, and the conductive projection **332** extending from an end of the conductive tubular member **332** so that a complete circuit is formed to ionize the argon gas for argon plasma coagulation. The ultrapolar electrosurgery blade assembly **300** of the present invention is also capable of cutting a patient's tissue using the sharp cutting edge (comprising conductive cutting ends **22**, **26** separated by sharp non-conductive cutting end **74**) of the ultrapolar electrosurgery blade **10** alone without any use of RF energy and without any use of argon plasma. The ultrapolar electrosurgery blade assembly **300** of the present invention can also enhance the cutting of a patient's tissue using the sharp conductive cutting edges **22**, **24** of the ultrapolar electrosurgery blade **10** by also supplying RF energy to the exposed portion of the active electrode **24** of the ultrapolar electrosurgery blade **10**. Moreover, the ultrapolar electrosurgery blade assembly **300** of the present invention having a sharp cutting edge and argon beam capability enables a user or surgeon to simultaneously perform cutting and

coagulation without the need to switch between cutting and coagulation modes by performing argon plasma assisted cutting and coagulation. For example, the sharp cutting edge of the ultrapolar electrosurgery blade **10** can be used without any RF energy for cutting while the conductive hollow tubular member **330** through which the argon gas is introduced, and which is contained within the non-conductive tube member **320**, is activated and directed via conductive projection **332** to provide ionized argon gas for argon plasma coagulation of tissue. In another example, low power may be applied to the ultrapolar electrosurgery blade **10** to coagulate tissue or enhance cutting of tissue while the conductive hollow tubular member **330** through which the argon gas is introduced, and which is contained within the non-conductive tube member **320**, is activated and directed via the conductive projection **332** to provide ionized argon gas for argon plasma coagulation of tissue.

(36) The drawings and description of exemplary embodiments of the invention herein shows various exemplary embodiments of the invention. These exemplary embodiments and modes are described in sufficient detail to enable those skilled in the art to practice the invention and are not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the following disclosure is intended to teach both the implementation of the exemplary embodiments and modes and any equivalent modes or embodiments that are known or obvious to those reasonably skilled in the art. Additionally, all included examples are non-limiting illustrations of the exemplary embodiments and modes, which similarly avail themselves to any equivalent modes or embodiments that are known or obvious to those reasonably skilled in the art.

(37) Other combinations and/or modifications of structures, arrangements, applications, proportions, elements, materials, or components used in the practice of the instant invention, in addition to those not specifically recited, can be varied or otherwise particularly adapted to specific environments, manufacturing specifications, design parameters, or other operating requirements without departing from the scope of the instant invention and are intended to be included in this disclosure.

(38) Unless specifically noted, it is the Applicant's intent that the words and phrases in the specification and the claims be given the commonly accepted generic meaning or an ordinary and accustomed meaning used by those of ordinary skill in the applicable arts. In the instance where these meanings differ, the words and phrases in the specification and the claims should be given the broadest possible, generic meaning. If any other special meaning is intended for any word or phrase, the specification will clearly state and define the special meaning.

Claims

1. An ultrapolar electrosurgery blade comprising: a top thin elongated conductive member and a bottom thin elongated conductive member in vertical alignment with one another and spaced apart from one another along their lengths wherein each of the top and bottom thin elongated conductive members includes opposing planar sides, a sharp distal-most cutting end, and a proximal non-cutting end; and a non-conductive housing or a non-conductive coating that covers the opposing planar sides of the top and bottom thin elongated conductive members in addition to at least a portion of a top of the top thin elongated conductive member and at least a portion of a bottom of the bottom thin elongated conductive member, and that fills the space between the top and bottom thin elongated conductive members wherein the non-conductive housing or non-conductive coating includes a sharp cutting end located between the sharp distal-most cutting ends of the top and bottom thin elongated conductive members and wherein at least a portion of the top of the top thin elongated conductive member is exposed between, and in a same continuous planar relationship with, top portions of the non-conductive housing or non-conductive coating located on a top of the ultrapolar electrosurgery blade.
2. The ultrapolar electrosurgery blade of claim 1 wherein at least a portion of the bottom of the bottom thin elongated conductive member is exposed in a same continuous planar relationship

with, portions of the non-conductive housing or non-conductive coating located on a bottom of the ultrapolar electrosurgery blade.

3. The ultrapolar electrosurgery blade of claim 1 wherein the top and bottom thin elongated conductive members comprise a hard metal.

4. The ultrapolar electrosurgery blade of claim 1 wherein the non-conductive housing or non-conductive coating comprises a ceramic material.

5. An ultrapolar electrosurgery blade assembly comprising the ultrapolar electrosurgery blade of claim 1 and a non-conductive tube member having a hollow tubular shaped opening contained therein and a slot wherein the slot is positioned over the top of the ultrapolar electrosurgery blade such that it is in direct physical contact with the ultrapolar electrosurgery blade.

6. The ultrapolar electrosurgery blade assembly of claim 5 further comprising a conductive hollow tubular member contained within at least a portion of the non-conductive tube member.

7. The ultrapolar electrosurgery blade assembly of claim 6 wherein the ultrapolar electrosurgery blade assembly further comprises a conductive projection extending from at least one of the conductive hollow tubular member contained within the non-conductive tube member and the portion of the top of the top thin elongated conductive member that is exposed between, and in a same continuous planar relationship with, top portions of the non-conductive housing or non-conductive coating located on top of the ultrapolar electrosurgery blade and contained within the non-conductive tube member.

8. The ultrapolar electrosurgery blade assembly of claim 6 wherein the non-conductive housing or non-conductive coating covers a portion of the top of the top thin elongated conductive member located between the conductive hollow tubular member and an exposed cutting end of the top thin elongated conductive member and the ultrapolar electrosurgery blade assembly further comprises a conductive projection extending from an end of the conductive hollow tubular member contained within the non-conductive tube member.

9. The ultrapolar electrosurgery blade assembly of claim 6 wherein the conductive hollow tubular member comprises a hard metal.

10. The ultrapolar electrosurgery blade assembly of claim 6 wherein the non-conductive tube member comprises a ceramic.

11. The ultrapolar electrosurgery blade assembly of claim 6 wherein the conductive hollow tubular member contained within the non-conductive tube member comprises a slot wherein the slot of the conductive hollow tubular member is also positioned over at least a portion of the top of the ultrapolar electrosurgery blade.

12. The ultrapolar electrosurgery blade assembly of claim 5 wherein the non-conductive housing or the non-conductive coating of the ultrapolar electrosurgery blade comprises a ceramic.

13. The ultrapolar electrosurgery blade assembly of claim 5 wherein the top and bottom thin elongated conductive members of the ultrapolar electrosurgery blade comprise a hard metal.

14. The ultrapolar electrosurgery blade of claim 1 further comprising a non-conductive support member having two openings therein in vertical alignment with one another wherein a portion of said top and bottom thin elongated conductive members are respectively contained within one of the two openings.

15. A method for making an ultrapolar electrosurgery blade comprising the steps of: providing a top thin elongated conductive member and a bottom thin elongated conductive member in vertical alignment with one another and spaced apart from one another along their lengths wherein each of the top and bottom thin elongated conductive members includes opposing planar sides, a sharp distal-most cutting end, and a proximal non-cutting end; and covering the opposing planar sides of the top and bottom thin elongated conductive members and at least a portion of a top of the top thin elongated conductive member and at least a portion of a bottom of the bottom thin elongated conductive member with a non-conductive housing or a non-conductive coating, wherein the non-conductive housing or non-conductive coating also fills the space between the top and bottom thin

elongated conductive members and includes a sharp cutting end located between the sharp distal-most cutting ends of the top and bottom thin elongated conductive members, and wherein at least a portion of the top of the top thin elongated conductive member is exposed between, and in a same continuous planar relationship with, top portions of the non-conductive housing or non-conductive coating located on a top of the ultrapolar electrosurgery blade.

16. The method of claim 15 wherein at least a portion of the bottom of the bottom thin elongated conductive member is exposed in a same continuous planar relationship with, portions of the non-conductive housing or non-conductive coating located on a bottom of the ultrapolar electrosurgery blade.

17. The method of claim 15 further comprising the step of positioning a non-conductive tube member having a hollow tubular shaped opening and a slot over the top of the ultrapolar electrosurgery blade to create an ultrapolar electrosurgery blade assembly wherein the slot is positioned over the top of the ultrapolar electrosurgery blade such that it is in direct physical contact with the ultrapolar electrosurgery blade.

18. The method of claim 17 further comprising the step of providing a conductive hollow tubular member such that it is contained within at least a portion of the non-conductive tube member.

19. The method of claim 18 further comprising the step of providing a conductive projection that extends from at least one of the conductive hollow tubular member contained within the non-conductive tube member and the portion of the top of the top thin elongated conductive member that is exposed between, and in a same continuous planar relationship with, top portions of the non-conductive housing or non-conductive coating located on top of the ultrapolar electrosurgery blade and contained within the non-conductive tube member.

20. The method of claim 15 further comprising the step of positioning a non-conductive support member having two openings therein in vertical alignment with one another such that a portion of said top and bottom thin elongated conductive members are respectively contained within one of the two openings.
